

Assessment for programming module-Copy8

January 9, 2025

1 Exploring Global Health Data

In this report, we will look at global health data, focussing on gender disparities and complete the following tasks:

- Initial Data Inspection
- Exploratory Data Analysis
- Statistical Analysis 1: How do lifestyle factors correlate with disease prevalence for males and for females?
- Statistical Analysis 2: Analysing the relationship between health behaviours and treatments and mortality risk from NCDs

1.1 Initial Data Inspection

1.1.1 Loading the Dataset and importing Modules

```
[56]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

The history saving thread hit an unexpected error (OperationalError('attempt to write a readonly database')).History will not be written to the database.

```
[57]: df = pd.read_csv("data/data_ncd.csv")
```

1.1.2 Exploring the Structure

We will start of by exploring the structure of the data to look at any potential issues with its quality. This will help us further on for data cleaning and data transformation.

```
[58]: df.head()
```

```
[58]:
```

	country	year	\
0	China	2018	
1	Georgia	2019	
2	Venezuela (Bolivarian Republic of)	2011	
3	Nepal	2019	
4	Bahamas	2019	

		factor	sex	estimate	\
0	Daily cigarette smoking among adults (%)		Male	44.0	
1	Daily cigarette smoking among adults (%)		Male	40.0	
2	Daily cigarette smoking among adults (%)		Male	17.0	
3	Daily cigarette smoking among adults (%)		Male	19.0	
4	Daily cigarette smoking among adults (%)		Male	12.0	

	wbincome2024
0	Upper middle income
1	Upper middle income
2	NaN
3	Lower middle income
4	High income

```
[59]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 110316 entries, 0 to 110315
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   country         110316 non-null object
1   year            110316 non-null int64
2   factor          110316 non-null object
3   sex             110316 non-null object
4   estimate        110310 non-null float64
5   wbincome2024    108682 non-null object
dtypes: float64(1), int64(1), object(4)
memory usage: 5.0+ MB
```

```
[60]: # Changing the 'year' data type to object
df['year'] = df['year'].astype('object')
# Summary Statistics for numerical column (estimates)
df.describe()
```

```
[60]:          estimate
count  110310.000000
mean    22.047394
std     16.252274
min      0.000000
25%      8.000000
50%     20.000000
75%     32.000000
max     88.000000
```

```
[61]: # Summary stats for objects
df.describe(include = object)
```

```
[61]:
```

	country	year	factor	\
count	110316	110316	110316	
unique	194	47	12	
top	Thailand	2015	Overweight prevalence among adults, BMI >= 25 ...	
freq	582	3760	15960	

	sex	wbincome2024
count	110316	108682
unique	2	4
top	Male	High income
freq	55158	34788

```
[62]: missing_data= df.isna().sum()
print('Missing Data:')
print(missing_data)
```

```
Missing Data:
country          0
year             0
factor           0
sex              0
estimate         6
wbincome2024    1634
dtype: int64
```

```
[63]: data_rows = df.shape[0]
data_columns = df.shape[1]
print(f'This dataset has {data_columns} columns and {data_rows} rows.')
```

This dataset has 6 columns and 110316 rows.

1.1.3 Information contained in the dataset

```
[64]: data_types = df.dtypes
data_types
```

```
[64]: country          object
year                object
factor             object
sex               object
estimate          float64
wbincome2024      object
dtype: object
```

Now that we know a little bit more about the structure of our data, we can also look more closely at what information it contains by looking at the unique values in the columns.

```
[65]: df["factor"].unique()
```

```
[65]: array(['Daily cigarette smoking among adults (%)',
        'Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%)',
        'Estimate of current cigarette smoking prevalence (%)',
        'Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%)',
        'Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%)',
        'Overweight prevalence among adults, BMI >= 25 (crude estimate) (%)',
        'Obesity prevalence among adults, BMI >= 30 (crude estimate) (%)',
        'Controlled hypertension among adults aged 30-79 years with hypertension (crude estimate) (%)',
        'Hypertension among adults aged 30-79 years (crude estimate) (%)',
        'Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (crude estimate) (%)',
        'Insufficient physical activity among adults aged 18+ years (crude estimate) (%)',
        'Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres of pure alcohol)'],
        dtype=object)
```

```
[66]: df['year'].unique()
```

```
[66]: array([2018, 2019, 2011, 2021, 2016, 2017, 2020, 2015, 2010, 2013, 2012,
        2009, 2008, 2014, 2001, 1976, 1989, 2005, 1983, 1988, 1986, 1996,
        1998, 1978, 1991, 1980, 1999, 2002, 1975, 1984, 1994, 1981, 2003,
        2004, 1995, 2006, 1977, 2007, 1985, 1979, 1982, 1992, 1987, 2000,
        1993, 1997, 1990], dtype=object)
```

```
[67]: df['country'].unique()
```

```
[67]: array(['China', 'Georgia', 'Venezuela (Bolivarian Republic of)', 'Nepal',
        'Bahamas', 'Republic of Moldova', 'Ethiopia', 'Armenia', 'Canada',
        'Mozambique', 'Egypt', 'Sudan', 'Austria', 'Costa Rica',
        'Lithuania', 'Serbia', 'Thailand', 'Eritrea', 'Czechia', 'Ecuador',
        'Bolivia (Plurinational State of)', 'Namibia', 'Equatorial Guinea',
        'Gabon', 'United Arab Emirates', 'Slovenia', 'Libya', 'Peru',
        'Seychelles', 'Botswana', 'Romania',
        'Micronesia (Federated States of)', 'Sri Lanka', 'Uzbekistan',
        'Turkmenistan', 'Cabo Verde', 'Republic of Korea',
        'Trinidad and Tobago', 'Saint Kitts and Nevis', 'Qatar',
        'Kyrgyzstan', 'Singapore', 'Kuwait', 'Mongolia', 'Nicaragua',
        'Saint Lucia', 'Niger', 'United Republic of Tanzania',
        'South Africa', 'Sao Tome and Principe', 'Guyana', 'Palau',
        'Honduras', 'Democratic Republic of the Congo', 'Suriname', 'Cuba',
        'Japan', 'Burundi', 'Portugal', 'Paraguay', 'Bangladesh',
        'Sierra Leone', 'Angola', 'Italy', 'Syrian Arab Republic',
        'Andorra', 'Tuvalu', 'Kiribati', 'Iceland', 'Cameroon',
```

```
'Azerbaijan', 'Bosnia and Herzegovina', 'Dominican Republic',
'Argentina', 'New Zealand', 'Nigeria', 'Cambodia', 'Bulgaria',
'Saint Vincent and the Grenadines', 'Somalia', 'Niue',
'Papua New Guinea', 'Morocco', 'Uganda', 'Estonia', 'Colombia',
'Mexico', 'Barbados', 'India', 'Iran (Islamic Republic of)',
'Denmark', 'Philippines', 'Vanuatu', 'Samoa',
'Lao People's Democratic Republic', 'Kenya', 'Mali',
'Burkina Faso', 'Antigua and Barbuda', 'Lebanon', 'Finland',
'Lesotho', 'Jordan', 'Zambia', 'Mauritania', 'Chile', 'Comoros',
'Slovakia', 'Montenegro', 'Türkiye', 'Croatia', 'Malta', 'France',
'Haiti', 'Liberia', 'Malaysia', 'Israel',
'Central African Republic', 'Solomon Islands', 'Ireland', 'Panama',
'Benin', 'Albania', 'Netherlands (Kingdom of the)', 'Viet Nam',
'Zimbabwe', 'Cyprus', 'Côte d'Ivoire', 'Yemen', 'Chad', 'Oman',
'Australia', 'Democratic People's Republic of Korea', 'Bhutan',
'Spain', 'Djibouti', 'Grenada', 'Nauru', 'Timor-Leste', 'Bahrain',
'Ukraine', 'The United Kingdom', 'Algeria', 'Greece',
'El Salvador', 'Maldives', 'Saudi Arabia', 'Brunei Darussalam',
'Cook Islands', 'Latvia', 'Ghana', 'Guatemala', 'Belize',
'Luxembourg', 'Belarus', 'Afghanistan', 'Iraq', 'Tajikistan',
'Kazakhstan', 'Guinea', 'Dominica', 'Togo', 'Guinea-Bissau',
'United States of America', 'Myanmar', 'Brazil', 'Congo',
'Madagascar', 'Belgium', 'Fiji', 'Poland', 'Uruguay',
'Switzerland', 'Eswatini', 'Hungary', 'Germany', 'Norway',
'Senegal', 'Pakistan', 'Tunisia', 'Marshall Islands', 'Indonesia',
'Rwanda', 'Jamaica', 'Sweden', 'Mauritius', 'Malawi', 'Gambia',
'Russian Federation', 'North Macedonia', 'Tonga', 'South Sudan',
'occupied Palestinian territory', 'Puerto Rico'], dtype=object)
```

1.2 Exploratory data analysis

1.2.1 Data cleaning and transformation

From the initial data analysis we could see that there are missing values in the income group and estimate columns. This data is also in a long format, but having a wide format where we can see each factor as a separate column would make it easier for us to analyse this data. We can also sort the rows out by country, year and sex. We also need to remove any duplicates (if there are any) and then handle the missing values in this dataset. We can also change the column names to make them easier to read.

```
[68]: # Count duplicates
counting_duplicates = df.duplicated().sum()
print(f'There are {counting_duplicates} duplicated rows')
```

There are 0 duplicated rows

```
[69]: # Viewing the rows containing the 6 missing values for 'estimate'
missing_estimates = df[df['estimate'].isna()]
```

```
missing_estimates
```

```
[69]:
```

	country	year	factor	sex	\
20	Mozambique	2011	Daily cigarette smoking among adults (%)	Female	
31	Gabon	2012	Daily cigarette smoking among adults (%)	Female	
34	Equatorial Guinea	2011	Daily cigarette smoking among adults (%)	Female	
35	Libya	2009	Daily cigarette smoking among adults (%)	Female	
66	Namibia	2013	Daily cigarette smoking among adults (%)	Female	
67	Nicaragua	2001	Daily cigarette smoking among adults (%)	Male	

	estimate	wbincome2024
20	NaN	Low income
31	NaN	Upper middle income
34	NaN	Upper middle income
35	NaN	Upper middle income
66	NaN	Upper middle income
67	NaN	Lower middle income

From the initial data analysis we could see that there are missing values in the income group and estimate columns. This data is also in a long format, but having a wide format where we can see each factor as a separate column would make it easier for us to analyse this data. Before we can pivot the data into a wide format we must first do hierarchical indexing. We can sort them by country and then year. Once we have done this we can remove any duplicates (if there are any) and then handle the missing values in this dataset. We can also change names of the new column headings and create new columns if needed for analysis.

All the missing values in the estimates is from the Daily cigarette smoking among adults (%). Five out of these missing values are for the females. No specific income group- missing values are from countries ranging from low income to upper middle class.

```
[70]: # Dropping the rows containing the missing values for Daily cigarette smoking_
      ↪among adults
df2 = df.dropna(subset = 'estimate')
# Check that the data now has no missing values in the 'estimate' column
df2.isna().sum()
```

```
[70]: country      0
      year        0
      factor      0
      sex         0
      estimate    0
      wbincome2024 1634
      dtype: int64
```

Now we have missing values only in the wbincome2024 column. Since each country has many rows of data, it may be possible to use the filled observation for one row of a country to fill the missing observation of another row for the same country. To do this it would be easier if indexed the data by country first.

```
[71]: # Checking again for duplicates
duplicate_rows = df[df.duplicated(subset=['country', 'year', 'sex', 'factor'],
    ↪keep=False)]
print("Duplicate rows in the original data:")
print(duplicate_rows)
```

Duplicate rows in the original data:
Empty DataFrame
Columns: [country, year, factor, sex, estimate, wbincome2024]
Index: []

```
[72]: factor_combinations = df.groupby(['country', 'year', 'sex'])['factor'].nunique()
factor_combinations
# From the output we can see the countries shown in the preview. There should
    ↪be 12 factors for each if complete data.
# do not have the same number of unique values and not 12 so combinations will
    ↪be missing- we expect some NaN to appear after pivoting.
```

```
[72]: country      year  sex
Afghanistan      1975  Female    4
                1975  Male      4
                1976  Female    4
                1976  Male      4
                1977  Female    4
                ..
occupied Palestinian territory  2017  Male      3
                                2018  Female    3
                                2018  Male      3
                                2019  Female    3
                                2019  Male      3

Name: factor, Length: 18000, dtype: int64
```

```
[73]: # Pivoting the data to a wide format- splitting 'factor' column
df_wide = df2.pivot_table(index=['country', 'year', 'sex'], columns='factor',
    ↪values='estimate', aggfunc='first')
df_wide.head()
```

```
[73]: factor      Alcohol, total per capita (15+) consumption (SDG
Indicator 3.5.2) (in litres of pure alcohol) \
country      year sex
Afghanistan  1975 Female      NaN
                Male      NaN
                1976 Female      NaN
                Male      NaN
                1977 Female      NaN

factor      Controlled hypertension among adults aged 30-79 years
with hypertension (crude estimate) (%) \
```

country	year	sex	
Afghanistan	1975	Female	NaN
		Male	NaN
	1976	Female	NaN
		Male	NaN
	1977	Female	NaN

factor		Daily cigarette smoking among adults (%) \
country	year sex	
Afghanistan	1975 Female	NaN
	Male	NaN
	1976 Female	NaN
	Male	NaN
	1977 Female	NaN

factor		Estimate of current cigarette smoking prevalence (%) \
country	year sex	
Afghanistan	1975 Female	NaN
	Male	NaN
	1976 Female	NaN
	Male	NaN
	1977 Female	NaN

factor		Hypertension among adults aged 30-79 years (crude estimate) (%) \
country	year sex	
Afghanistan	1975 Female	NaN
	Male	NaN
	1976 Female	NaN
	Male	NaN
	1977 Female	NaN

factor		Insufficient physical activity among adults aged 18+ years (crude estimate) (%) \
country	year sex	
Afghanistan	1975 Female	NaN
	Male	NaN
	1976 Female	NaN
	Male	NaN
	1977 Female	NaN

factor		Obesity prevalence among adults, BMI >= 30 (crude estimate) (%) \
country	year sex	
Afghanistan	1975 Female	1.0
	Male	0.0
	1976 Female	1.0

	Male	0.0
	1977 Female	1.0

factor Overweight prevalence among adults, BMI >= 25 (crude estimate) (%) \

country	year	sex	
Afghanistan	1975	Female	6.0
		Male	5.0
	1976	Female	6.0
		Male	5.0
	1977	Female	7.0

factor Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%) \

country	year	sex	
Afghanistan	1975	Female	NaN
		Male	NaN
	1976	Female	NaN
		Male	NaN
	1977	Female	NaN

factor Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%) \

country	year	sex	
Afghanistan	1975	Female	17.0
		Male	20.0
	1976	Female	17.0
		Male	20.0
	1977	Female	17.0

factor Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (crude estimate) (%) \

country	year	sex	
Afghanistan	1975	Female	NaN
		Male	NaN
	1976	Female	NaN
		Male	NaN
	1977	Female	NaN

factor Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%)

country	year	sex	
Afghanistan	1975	Female	28.0
		Male	27.0
	1976	Female	28.0
		Male	27.0

```
[75]: # Resetting index so we can work with all the columns
df_reset = df_wide.reset_index()

# Merging data to include the columns with income groups (wbincome2024) and
↳handling missing income groups
keep_columns = df2[['country', 'year', 'sex', 'wbincome2024']]
df_full = pd.merge(df_reset, keep_columns, on=['country', 'year', 'sex'],
↳how='left')
```

```
[76]: # Handling the income group missing values
df_full['wbincome2024'] = df_full.groupby('country')['wbincome2024'].ffill().
↳bfill()
df_full.head()
```

```
[76]:      country  year  sex \
0  Afghanistan  1975  Female
1  Afghanistan  1975  Female
2  Afghanistan  1975  Female
3  Afghanistan  1975  Female
4  Afghanistan  1975   Male
```

Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres of pure alcohol) \

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

Controlled hypertension among adults aged 30-79 years with hypertension (crude estimate) (%) \

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

Daily cigarette smoking among adults (%) \

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

Estimate of current cigarette smoking prevalence (%) \

0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Hypertension among adults aged 30-79 years (crude estimate) (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Insufficient physical activity among adults aged 18+ years (crude estimate) (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Obesity prevalence among adults, BMI >= 30 (crude estimate) (%) \	
0	1.0
1	1.0
2	1.0
3	1.0
4	0.0

Overweight prevalence among adults, BMI >= 25 (crude estimate) (%) \	
0	6.0
1	6.0
2	6.0
3	6.0
4	5.0

Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%) \	
0	17.0
1	17.0

2	17.0
3	17.0
4	20.0

Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (crude estimate) (%) \

0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%) \

0	28.0
1	28.0
2	28.0
3	28.0
4	27.0

wbincome2024

0	Low income
1	Low income
2	Low income
3	Low income
4	Low income

```
[77]: # Checking that the income group has been filled with the correct countries.
check_income_fill = df_full.groupby('country')['wbincome2024'].nunique()
check_income_fill
```

```
[77]: country
Afghanistan      1
Albania          1
Algeria          1
Andorra          1
Angola           1
..
Viet Nam         1
Yemen            1
Zambia           1
Zimbabwe         1
occupied Palestinian territory  1
Name: wbincome2024, Length: 194, dtype: int64
```

```
[78]: # Number of rows and columns
columns_full = df_full.shape[1]
rows_full = df_full.shape[0]
```

```
print(f' This has {rows_full} rows and {columns_full} columns')
```

This has 110310 rows and 16 columns

```
[79]: # Checking the missing values in each column for the factors
missing_df_full = df_full.isna().sum()
print('Missing Data after splitting factors column:')
print(missing_df_full)
```

Missing Data after splitting factors column:

country

0

year

0

sex

0

Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres of pure alcohol) 93666

Controlled hypertension among adults aged 30-79 years with hypertension (crude estimate) (%) 23488

Daily cigarette smoking among adults (%) 109764

Estimate of current cigarette smoking prevalence (%) 93534

Hypertension among adults aged 30-79 years (crude estimate) (%) 23488

Insufficient physical activity among adults aged 18+ years (crude estimate) (%) 107748

Obesity prevalence among adults, BMI >= 30 (crude estimate) (%) 6416

Overweight prevalence among adults, BMI >= 25 (crude estimate) (%) 6416

Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%) 53004

Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%) 9386

Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (crude estimate) (%) 23488

Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%) 6416

wbincome2024

0

dtype: int64

```
[80]: # Proportion of missing data in each column
missing_proportion = df_full.isna().mean()
missing_proportion
```

```

[80]: country
      0.000000
      year
      0.000000
      sex
      0.000000
      Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres of
      pure alcohol)                                0.849116
      Controlled hypertension among adults aged 30-79 years with hypertension (crude
      estimate) (%)                                0.212927
      Daily cigarette smoking among adults (%)
      0.995050
      Estimate of current cigarette smoking prevalence (%)
      0.847919
      Hypertension among adults aged 30-79 years (crude estimate) (%)
      0.212927
      Insufficient physical activity among adults aged 18+ years (crude estimate) (%)
      0.976775
      Obesity prevalence among adults, BMI >= 30 (crude estimate) (%)
      0.058163
      Overweight prevalence among adults, BMI >= 25 (crude estimate) (%)
      0.058163
      Probability of dying between age 30 and exact age 70 from any of cardiovascular
      disease, cancer, diabetes or chronic respiratory disease (%)    0.480500
      Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%)
      0.085087
      Treatment (taking medicine) for hypertension among adults aged 30-79 with
      hypertension (crude estimate) (%)                                0.212927
      Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%)
      0.058163
      wbincome2024
      0.000000
      dtype: float64

```

There is a considerable amount of missing data- 10 out of 12 factor columns contain missing data with some columns having almost all of the data missing. Particularly for the columns for the Daily Smoking prevalence, insufficient physical activity and alcohol total per capita. We are however tasked to explore lifestyle factors and health behaviours in our analysis and there are only 4 columns containing such information. Removing these columns would mean we would only have 1 lifestyle factor to analyse- and even that column (estimate of current cigarette smoking prevalence) contains 15690 missing values (87% of the data in the column). For this reason I will remove the columns with over 90% missing and use mean imputation by sex and country to fill in the remaining missing data for the remaining 7 columns with missing data. We do need to keep in mind that this may impact the reliability of our results when we come to analyse our data.

```
[81]: # Removing the 3 columns containing more than 89.5 % of the data missing
df_dropped_columns = df_full.drop(columns=missing_proportion[missing_proportion_
↪ > 0.895].index)
df_dropped_columns.head()
```

```
[81]:      country  year  sex \
0  Afghanistan  1975  Female
1  Afghanistan  1975  Female
2  Afghanistan  1975  Female
3  Afghanistan  1975  Female
4  Afghanistan  1975   Male
```

```
      Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres
of pure alcohol) \
```

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

```
      Controlled hypertension among adults aged 30-79 years with hypertension
(crude estimate) (%) \
```

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

```
      Estimate of current cigarette smoking prevalence (%) \
```

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

```
      Hypertension among adults aged 30-79 years (crude estimate) (%) \
```

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

```
      Obesity prevalence among adults, BMI >= 30 (crude estimate) (%) \
```

```
0      1.0
1      1.0
2      1.0
3      1.0
```

4	0.0
Overweight prevalence among adults, BMI >= 25 (crude estimate) (%) \	
0	6.0
1	6.0
2	6.0
3	6.0
4	5.0
Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (crude estimate) (%) \	
0	17.0
1	17.0
2	17.0
3	17.0
4	20.0
Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (crude estimate) (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
Underweight prevalence among adults, BMI < 18.5 (crude estimate) (%) \	
0	28.0
1	28.0
2	28.0
3	28.0
4	27.0
wbincome2024	
0	Low income
1	Low income
2	Low income
3	Low income
4	Low income


```
[82]: # Cleaning the column headings- using string manipulation
df_dropped_columns.columns = [col.replace(' (crude estimate)', '') for col in df_dropped_columns.columns]
df_dropped_columns.head()
```

```
[82]:
```

	country	year	sex	\
0	Afghanistan	1975	Female	
1	Afghanistan	1975	Female	
2	Afghanistan	1975	Female	
3	Afghanistan	1975	Female	
4	Afghanistan	1975	Male	

	Alcohol, total per capita (15+) consumption (SDG Indicator 3.5.2) (in litres of pure alcohol)	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

	Controlled hypertension among adults aged 30-79 years with hypertension (%)	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

	Estimate of current cigarette smoking prevalence (%)	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

	Hypertension among adults aged 30-79 years (%)	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

	Obesity prevalence among adults, BMI >= 30 (%)	\
0	1.0	
1	1.0	
2	1.0	
3	1.0	

4	0.0
---	-----

Overweight prevalence among adults, BMI >= 25 (%) \	
0	6.0
1	6.0
2	6.0
3	6.0
4	5.0

Probability of dying between age 30 and exact age 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory disease (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years) (%) \	
0	17.0
1	17.0
2	17.0
3	17.0
4	20.0

Treatment (taking medicine) for hypertension among adults aged 30-79 with hypertension (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Underweight prevalence among adults, BMI < 18.5 (%) wbincome2024		
0	28.0	Low income
1	28.0	Low income
2	28.0	Low income
3	28.0	Low income
4	27.0	Low income

```
[83]: # Further amending column headings
new_column_names = {'Alcohol, total per capita (15+) consumption (SDG Indicator_
↳3.5.2) (in litres of pure alcohol)': 'Alcohol Consumption',
                    'Probability of dying between age 30 and exact age 70 from_
↳any of cardiovascular disease, cancer, diabetes or chronic respiratory_
↳disease (%)': 'Probability of death from non-communicable diseases (%)',
                    'Treatment (taking medicine) for hypertension among adults_
↳aged 30-79 with hypertension (%)': 'Receiving hypertension treatment (%)',
```

```

        'wbincome2024': 'Income Group',
        'country': 'Country',
        'year': 'Year',
        'sex': 'Sex',
        'Controlled hypertension among adults aged 30-79 years with
↳hypertension (%)': 'Controlled hypertension (%)',
        'Raised blood pressure (SBP>=140 OR DBP>=90) (18+ years)
↳(%))': 'Raised Blood pressure (%)',
        'Obesity prevalence among adults, BMI >= 30 (%)': 'Obesity
↳prevalence among adults (%)',
        'Underweight prevalence among adults, BMI < 18.5 (%)' :
↳'Underweight prevalence among adults (%)',
        'Overweight prevalence among adults, BMI >= 25 (%)':
↳'Overweight prevalence among adults(%)'}

df_full = df_dropped_columns.rename(columns=new_column_names)
df_full.head()

```

```

[83]:
      Country  Year  Sex  Alcohol Consumption \
0  Afghanistan  1975  Female                NaN
1  Afghanistan  1975  Female                NaN
2  Afghanistan  1975  Female                NaN
3  Afghanistan  1975  Female                NaN
4  Afghanistan  1975  Male                  NaN

      Controlled hypertension (%) \
0                NaN
1                NaN
2                NaN
3                NaN
4                NaN

      Estimate of current cigarette smoking prevalence (%) \
0                NaN
1                NaN
2                NaN
3                NaN
4                NaN

      Hypertension among adults aged 30-79 years (%) \
0                NaN
1                NaN
2                NaN
3                NaN
4                NaN

      Obesity prevalence among adults (%)  Overweight prevalence among adults(%) \

```

0	1.0	6.0
1	1.0	6.0
2	1.0	6.0
3	1.0	6.0
4	0.0	5.0

Probability of death from non-communicable diseases (%) \	
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

Raised Blood pressure (%)		Receiving hypertension treatment (%) \	
0	17.0		NaN
1	17.0		NaN
2	17.0		NaN
3	17.0		NaN
4	20.0		NaN

Underweight prevalence among adults (%)		Income Group	
0	28.0	Low	income
1	28.0	Low	income
2	28.0	Low	income
3	28.0	Low	income
4	27.0	Low	income

Checking distributions to decide between mean and median imputation for the data

We have already removed the columns where there was too much missing data. We now need to handle the missing values in the remaining columns by either mean or median imputation. We can look at the distribution of the variables to decide what to use.

```
[84]: # Including nan rows
summ_stats_with_nan= df_full.describe()
summ_stats_with_nan
```

```
[84]:      Alcohol Consumption  Controlled hypertension (%) \
count      16644.000000      86822.000000
mean         5.611512         11.067218
std         5.956350         9.620022
min          0.000000         1.000000
25%          1.000000         4.000000
50%          3.000000         8.000000
75%          8.000000        15.000000
max         30.000000        67.000000
```

Estimate of current cigarette smoking prevalence (%) \	
--	--

count	16776.000000
mean	19.477825
std	15.260062
min	0.000000
25%	5.000000
50%	18.000000
75%	29.000000
max	74.000000

Hypertension among adults aged 30-79 years (%) \	
count	86822.000000
mean	36.426781
std	7.940251
min	18.000000
25%	31.000000
50%	35.000000
75%	41.000000
max	61.000000

Obesity prevalence among adults (%) \	
count	103894.000000
mean	13.063324
std	10.746784
min	0.000000
25%	4.000000
50%	11.000000
75%	20.000000
max	63.000000

Overweight prevalence among adults(%) \	
count	103894.000000
mean	38.255337
std	19.164162
min	3.000000
25%	21.000000
50%	40.000000
75%	54.000000
max	88.000000

Probability of death from non-communicable diseases (%) \	
count	57306.000000
mean	22.320734
std	9.418436
min	4.000000
25%	15.000000
50%	21.000000
75%	28.000000

max 64.000000

	Raised Blood pressure (%)	Receiving hypertension treatment (%) \
count	100924.000000	86822.000000
mean	24.901569	30.733789
std	6.885457	17.274786
min	8.000000	3.000000
25%	20.000000	16.000000
50%	23.000000	28.000000
75%	28.000000	43.000000
max	52.000000	80.000000

	Underweight prevalence among adults (%)
count	103894.000000
mean	7.579995
std	6.691297
min	0.000000
25%	2.000000
50%	5.000000
75%	12.000000
max	37.000000

```
[85]: # Without nan rows
summ_stats_no_nan = df_full.dropna().describe()
summ_stats_no_nan
```

	Alcohol Consumption	Controlled hypertension (%) \
count	12642.000000	12642.000000
mean	5.835074	13.135105
std	6.153645	10.391556
min	0.000000	1.000000
25%	1.000000	5.000000
50%	4.000000	10.000000
75%	9.000000	18.000000
max	30.000000	65.000000

	Estimate of current cigarette smoking prevalence (%) \
count	12642.000000
mean	19.236592
std	15.313209
min	0.000000
25%	5.000000
50%	18.000000
75%	29.000000
max	74.000000

Hypertension among adults aged 30-79 years (%) \

count	12642.000000
mean	36.644281
std	8.220602
min	18.000000
25%	31.000000
50%	35.000000
75%	42.000000
max	61.000000

	Obesity prevalence among adults (%) \
count	12642.000000
mean	15.139218
std	10.134000
min	0.000000
25%	5.000000
50%	15.000000
75%	22.000000
max	53.000000

	Overweight prevalence among adults(%) \
count	12642.000000
mean	42.427780
std	18.588569
min	7.000000
25%	25.000000
50%	47.000000
75%	58.000000
max	80.000000

	Probability of death from non-communicable diseases (%) \
count	12642.000000
mean	22.140484
std	9.498569
min	5.000000
25%	15.000000
50%	21.000000
75%	28.000000
max	64.000000

	Raised Blood pressure (%)	Receiving hypertension treatment (%) \
count	12642.000000	12642.000000
mean	24.086853	34.173548
std	6.198051	17.809443
min	8.000000	4.000000
25%	20.000000	19.000000
50%	23.000000	33.000000
75%	27.000000	48.000000

max	45.000000	78.000000
-----	-----------	-----------

Underweight prevalence among adults (%)	
count	12642.000000
mean	6.501582
std	5.984650
min	0.000000
25%	2.000000
50%	4.000000
75%	11.000000
max	30.000000

```
[86]: # Distribution without missing values (after dropping missing values)

#Separating by males and females
female_data= df_full[df_full['Sex'] == 'Female'].dropna()
male_data= df_full[df_full['Sex'] == 'Male'].dropna()

# Setting the colours
colors = sns.color_palette('tab20c')

# figure and axes for subplots
fig, axes = plt.subplots(4, 3, figsize=(18, 16))

# Overlaying Histogram 1
sns.histplot(female_data['Controlled hypertension (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[0, 0],
             label='Female', stat='density')
sns.histplot(male_data['Controlled hypertension (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[0, 0], label='Male',
             stat='density')
axes[0, 0].set_title('Controlled Hypertension')
axes[0, 0].set_xlabel('Controlled Hypertension %')
axes[0, 0].set_ylabel('Frequency')
axes[0, 0].legend()

# Overlaying Histogram 2
sns.histplot(female_data['Estimate of current cigarette smoking prevalence (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[0, 1],
             label='Female', stat='density')
sns.histplot(male_data['Estimate of current cigarette smoking prevalence (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[0, 1], label='Male',
             stat='density')
axes[0, 1].set_title('Cigarette Smoking Prevalence')
axes[0, 1].set_xlabel('Cigarette Smoking Prevalence %')
axes[0, 1].set_ylabel('Frequency')
```



```

axes[0, 1].legend()

# Overlaying Histogram 3
sns.histplot(female_data['Hypertension among adults aged 30-79 years (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[0, 2],
             label='Female', stat='density')
sns.histplot(male_data['Hypertension among adults aged 30-79 years (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[0, 2], label='Male',
             stat='density')
axes[0, 2].set_title('Hypertension Prevalence')
axes[0, 2].set_xlabel('Hypertension Prevalence %')
axes[0, 2].set_ylabel('Frequency')
axes[0, 2].legend()

# Overlaying Histogram 4
sns.histplot(female_data['Obesity prevalence among adults (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[1, 0],
             label='Female', stat='density')
sns.histplot(male_data['Obesity prevalence among adults (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[1, 0], label='Male',
             stat='density')
axes[1, 0].set_title('Obesity Prevalence')
axes[1, 0].set_xlabel('Obesity Prevalence %')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].legend()

# Overlaying Histogram 5
sns.histplot(female_data['Overweight prevalence among adults(%)'],
             bins=30, color=colors[16], kde=False, ax=axes[1, 1],
             label='Female', stat='density')
sns.histplot(male_data['Overweight prevalence among adults(%)'],
             bins=30, color=colors[4], kde=False, ax=axes[1, 1], label='Male',
             stat='density')
axes[1, 1].set_title('Overweight Prevalence')
axes[1, 1].set_xlabel('Overweight Prevalence %')
axes[1, 1].set_ylabel('Frequency')
axes[1, 1].legend()

# Overlaying Histogram 6
sns.histplot(female_data['Probability of death from non-communicable diseases_
↳(%)'],
             bins=30, color=colors[16], kde=False, ax=axes[1, 2],
             label='Female', stat='density')
sns.histplot(male_data['Probability of death from non-communicable diseases_
↳(%)'],

```

```

        bins=30, color=colors[4], kde=False, ax=axes[1, 2], label='Male',
        stat='density')
axes[1, 2].set_title('Probability of Dying 30-70 from NCDs')
axes[1, 2].set_xlabel('Probability of Dying 30-70 from NCDs')
axes[1, 2].set_ylabel('Frequency')
axes[1, 2].legend()

# Overlaying Histogram 7
sns.histplot(female_data['Raised Blood pressure (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[2, 0],
             label='Female', stat='density')
sns.histplot(male_data['Raised Blood pressure (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[2, 0], label='Male',
             stat='density')
axes[2, 0].set_title('Raised Blood Pressure')
axes[2, 0].set_xlabel('Raised Blood Pressure %')
axes[2, 0].set_ylabel('Frequency')
axes[2, 0].legend()

# Overlaying Histogram 8
sns.histplot(female_data['Receiving hypertension treatment (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[2, 1],
             label='Female', stat='density')
sns.histplot(male_data['Receiving hypertension treatment (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[2, 1], label='Male',
             stat='density')
axes[2, 1].set_title('Treatment for Hypertension')
axes[2, 1].set_xlabel('Treatment for Hypertension %')
axes[2, 1].set_ylabel('Frequency')
axes[2, 1].legend()

# Overlaying Histogram 9
sns.histplot(female_data['Underweight prevalence among adults (%)'],
             bins=30, color=colors[16], kde=False, ax=axes[2, 2],
             label='Female', stat='density')
sns.histplot(male_data['Underweight prevalence among adults (%)'],
             bins=30, color=colors[4], kde=False, ax=axes[2, 2], label='Male',
             stat='density')
axes[2, 2].set_title('Underweight Prevalence')
axes[2, 2].set_xlabel('Underweight Prevalence %')
axes[2, 2].set_ylabel('Frequency')
axes[2, 2].legend()

# Overlaying Histogram 10
sns.histplot(female_data['Alcohol Consumption'],

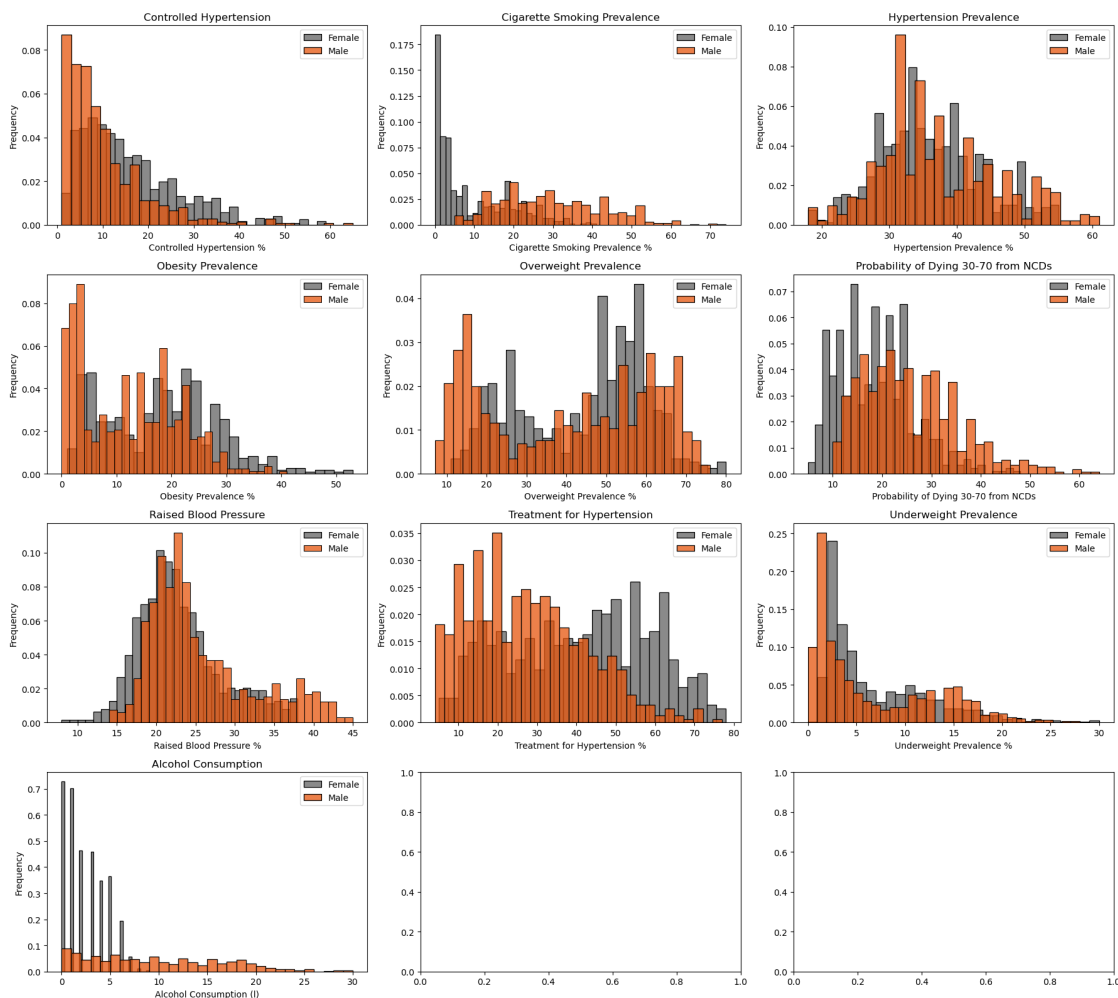
```

```

        bins=30, color=colors[16], kde=False, ax=axes[3, 0],
        label='Female', stat='density')
sns.histplot(male_data['Alcohol Consumption'],
             bins=30, color=colors[4], kde=False, ax=axes[3, 0], label='Male',
             stat='density')
axes[3, 0].set_title('Alcohol Consumption')
axes[3, 0].set_xlabel('Alcohol Consumption (l)')
axes[3, 0].set_ylabel('Frequency')
axes[3, 0].legend()

# Show the plots
plt.tight_layout()
plt.show()

```



```

[88]: # Using median imputation by sex
df_filled = df_full.groupby('Sex',

```

```

                                group_keys=False).apply(lambda group: group.
↪fillna(group.median(numeric_only=True)))

```

```

/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/3042316099.py:3
: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns.
This behavior is deprecated, and in a future version of pandas the grouping
columns will be excluded from the operation. Either pass `include_groups=False`
to exclude the groupings or explicitly select the grouping columns after groupby
to silence this warning.

```

```

    group_keys=False).apply(lambda group:
group.fillna(group.median(numeric_only=True)))

```

```

[89]: # Viewing the data
df_filled.head()

```

```

[89]:      Country  Year  Sex  Alcohol Consumption  \
0  Afghanistan  1975  Female                2.0
1  Afghanistan  1975  Female                2.0
2  Afghanistan  1975  Female                2.0
3  Afghanistan  1975  Female                2.0
4  Afghanistan  1975   Male                8.0

      Controlled hypertension (%)  \
0                      11.0
1                      11.0
2                      11.0
3                      11.0
4                      6.0

      Estimate of current cigarette smoking prevalence (%)  \
0                      5.0
1                      5.0
2                      5.0
3                      5.0
4                     28.0

      Hypertension among adults aged 30-79 years (%)  \
0                      35.0
1                      35.0
2                      35.0
3                      35.0
4                      36.0

      Obesity prevalence among adults (%)  Overweight prevalence among adults(%)  \
0                      1.0                      6.0
1                      1.0                      6.0
2                      1.0                      6.0
3                      1.0                      6.0

```

4	0.0	5.0
---	-----	-----

	Probability of death from non-communicable diseases (%) \
0	18.0
1	18.0
2	18.0
3	18.0
4	25.0

	Raised Blood pressure (%)	Receiving hypertension treatment (%) \
0	17.0	37.0
1	17.0	37.0
2	17.0	37.0
3	17.0	37.0
4	20.0	22.0

	Underweight prevalence among adults (%)	Income Group
0	28.0	Low income
1	28.0	Low income
2	28.0	Low income
3	28.0	Low income
4	27.0	Low income

```
[90]: # Check again that there are no missing values in this dataset
missing_data_after = df_filled.isna().sum()
missing_data_after
```

```
[90]: Country                                0
Year                                          0
Sex                                           0
Alcohol Consumption                         0
Controlled hypertension (%)                 0
Estimate of current cigarette smoking prevalence (%) 0
Hypertension among adults aged 30-79 years (%) 0
Obesity prevalence among adults (%)         0
Overweight prevalence among adults(%)       0
Probability of death from non-communicable diseases (%) 0
Raised Blood pressure (%)                   0
Receiving hypertension treatment (%)        0
Underweight prevalence among adults (%)     0
Income Group                               0
dtype: int64
```

```
[91]: data = df_filled
data.head()
```

[91]:

	Country	Year	Sex	Alcohol Consumption	\
0	Afghanistan	1975	Female	2.0	
1	Afghanistan	1975	Female	2.0	
2	Afghanistan	1975	Female	2.0	
3	Afghanistan	1975	Female	2.0	
4	Afghanistan	1975	Male	8.0	

	Controlled hypertension (%)	\
0	11.0	
1	11.0	
2	11.0	
3	11.0	
4	6.0	

	Estimate of current cigarette smoking prevalence (%)	\
0	5.0	
1	5.0	
2	5.0	
3	5.0	
4	28.0	

	Hypertension among adults aged 30-79 years (%)	\
0	35.0	
1	35.0	
2	35.0	
3	35.0	
4	36.0	

	Obesity prevalence among adults (%)	Overweight prevalence among adults(%)	\
0	1.0	6.0	
1	1.0	6.0	
2	1.0	6.0	
3	1.0	6.0	
4	0.0	5.0	

	Probability of death from non-communicable diseases (%)	\
0	18.0	
1	18.0	
2	18.0	
3	18.0	
4	25.0	

	Raised Blood pressure (%)	Receiving hypertension treatment (%)	\
0	17.0	37.0	
1	17.0	37.0	
2	17.0	37.0	
3	17.0	37.0	

4	20.0	22.0
---	------	------

	Underweight prevalence among adults (%)	Income Group
0	28.0	Low income
1	28.0	Low income
2	28.0	Low income
3	28.0	Low income
4	27.0	Low income

1.2.2 Looking more closely at Gender Disparities for the different health estimates

Now that the data has been cleaned and transformed, we can look at the topic of interest for our analysis- gender disparity.

```
[100]: # Summary Statistics by 'Sex'
summary_statistics = data.groupby('Sex').describe()
summary_statistics
```

```
[100]:      Alcohol Consumption
      count      mean      std  min  25%  50%  75%  max \
Sex
Female    55153.0  2.041847  0.762972  0.0  2.0  2.0  2.0  9.0
Male      55157.0  8.142684  2.625462  0.0  8.0  8.0  8.0  30.0
```

```
      Controlled hypertension (%)
      count      mean  ... \
Sex
Female    55153.0  13.098798  ...
Male      55157.0   7.942564  ...
```

```
      Receiving hypertension treatment (%)
      75%  max \
Sex
Female    47.0  80.0
Male      29.0  79.0
```

```
      Underweight prevalence among adults (%)
      count      mean      std  min  25% \
Sex
Female    55153.0  7.421808  6.079129  0.0  3.0
Male      55157.0  7.438059  6.936297  0.0  1.0
```

```
      50%  75%  max
Sex
Female  5.0  11.0  37.0
Male    5.0  13.0  35.0
```

[2 rows x 80 columns]

```
[93]: # Looking at boxplots to compare the distributions

# figure and axes
fig, axes = plt.subplots(4, 3, figsize=(30, 18))

# Boxplot 1
sns.boxplot(x='Sex', y='Controlled hypertension (%)', data=data, ax=axes[0, 0],
            palette=[colors[16], colors[4]])
axes[0, 0].set_title('Controlled Hypertension')
axes[0, 0].set_xlabel('Sex')
axes[0, 0].set_ylabel('Controlled Hypertension %')

# Boxplot 2
sns.boxplot(x='Sex', y='Estimate of current cigarette smoking prevalence (%)',
            data=data, ax=axes[0, 1], palette=[colors[16], colors[4]])
axes[0, 1].set_title('Cigarette Smoking Prevalence')
axes[0, 1].set_xlabel('Sex')
axes[0, 1].set_ylabel('Cigarette Smoking Prevalence %')

# Boxplot 3
sns.boxplot(x='Sex', y='Hypertension among adults aged 30-79 years (%)',
            data=data, ax=axes[0, 2], palette=[colors[16], colors[4]])
axes[0, 2].set_title('Hypertension Prevalence')
axes[0, 2].set_xlabel('Sex')
axes[0, 2].set_ylabel('Hypertension Prevalence %')

# Boxplot 4
sns.boxplot(x='Sex', y='Obesity prevalence among adults (%)', data=data,
            ax=axes[1, 0], palette=[colors[16], colors[4]])
axes[1, 0].set_title('Obesity Prevalence')
axes[1, 0].set_xlabel('Sex')
axes[1, 0].set_ylabel('Obesity Prevalence %')

# Boxplot 5
sns.boxplot(x='Sex', y='Overweight prevalence among adults(%)', data=data,
            ax=axes[1, 1], palette=[colors[16], colors[4]])
axes[1, 1].set_title('Overweight Prevalence')
axes[1, 1].set_xlabel('Sex')
axes[1, 1].set_ylabel('Overweight Prevalence %')

# Boxplot 6
sns.boxplot(x='Sex', y='Probability of death from non-communicable diseases,
            (%)', data=data, ax=axes[1, 2], palette=[colors[16], colors[4]])
axes[1, 2].set_title('Probability of Dying 30-70 from NCDs')
```



```

axes[1, 2].set_xlabel('Sex')
axes[1, 2].set_ylabel('Probability of Dying 30-70 from NCDs')

# Boxplot 7
sns.boxplot(x='Sex', y='Raised Blood pressure (%)', data=data, ax=axes[2, 0],
            palette=[colors[16], colors[4]])
axes[2, 0].set_title('Raised Blood Pressure')
axes[2, 0].set_xlabel('Sex')
axes[2, 0].set_ylabel('Raised Blood Pressure %')

# Boxplot 8
sns.boxplot(x='Sex', y='Receiving hypertension treatment (%)', data=data,
            ax=axes[2, 1], palette=[colors[16], colors[4]])
axes[2, 1].set_title('Treatment for Hypertension')
axes[2, 1].set_xlabel('Sex')
axes[2, 1].set_ylabel('Treatment for Hypertension %')

# Boxplot 9
sns.boxplot(x='Sex', y='Underweight prevalence among adults (%)', data=data,
            ax=axes[2, 2], palette=[colors[16], colors[4]])
axes[2, 2].set_title('Underweight Prevalence')
axes[2, 2].set_xlabel('Sex')
axes[2, 2].set_ylabel('Underweight Prevalence %')

# Boxplot 10 (Alcohol factor)
sns.boxplot(x='Sex', y='Alcohol Consumption', data=data, ax=axes[3, 0],
            palette=[colors[16], colors[4]])
axes[3, 0].set_title('Alcohol Consumption')
axes[3, 0].set_xlabel('Sex')
axes[3, 0].set_ylabel('Alcohol Consumption (l)')

plt.tight_layout()
plt.show()

```

/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:8
: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```

sns.boxplot(x='Sex', y='Controlled hypertension (%)', data=data, ax=axes[0,
0], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:1
4: FutureWarning:

```

Passing `palette` without assigning `hue` is deprecated and will be removed in

v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Estimate of current cigarette smoking prevalence (%)',
data=data, ax=axes[0, 1], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:2
0: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Hypertension among adults aged 30-79 years (%)',
data=data, ax=axes[0, 2], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:2
6: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Obesity prevalence among adults (%)', data=data,
ax=axes[1, 0], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:3
2: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Overweight prevalence among adults (%)', data=data,
ax=axes[1, 1], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:3
8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Probability of death from non-communicable diseases
(%)', data=data, ax=axes[1, 2], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:4
4: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Raised Blood pressure (%)', data=data, ax=axes[2, 0],
palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:5
0: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

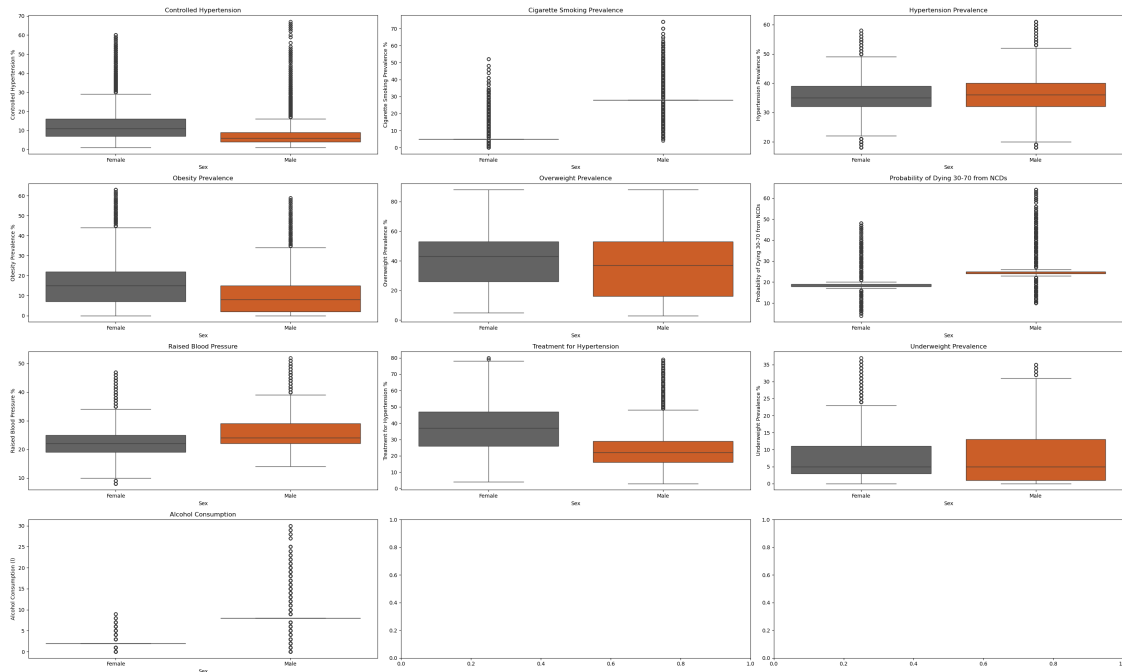
```
sns.boxplot(x='Sex', y='Receiving hypertension treatment (%)', data=data,
ax=axes[2, 1], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:5
6: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Underweight prevalence among adults (%)', data=data,
ax=axes[2, 2], palette=[colors[16], colors[4]])
/var/folders/vz/57y5gz5j22q26sncy4063qv80000gn/T/ipykernel_65653/2678157665.py:6
2: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Sex', y='Alcohol Consumption', data=data, ax=axes[3, 0],
palette=[colors[16], colors[4]])
```



Visual analysis for boxplot distributions For controlled hypertension, obesity prevalence, overweight prevalence and treatment for hypertension the median for women was higher than the median for men.

For Alcohol consumption, cigarette smoking and the probability of dying from non-communicable disease the median for Men was higher. However for these factors, we had a lot of missing values in the original data which we imputed with the median. This likely explains the extremely small interquartile ranges- most of the data was the same value (the median by sex).

Raised blood pressure also had a higher median but the interquartile range for this was not as small- less missing values in the original data column.

```
[94]: # list of all 10 factors
all_factors = [
    'Controlled hypertension (%)',
    'Hypertension among adults aged 30-79 years (%)',
    'Raised Blood pressure (%)',
    'Obesity prevalence among adults (%)',
    'Overweight prevalence among adults(%)',
    'Estimate of current cigarette smoking prevalence (%)',
    'Alcohol Consumption',
    'Receiving hypertension treatment (%)',
    'Underweight prevalence among adults (%)',
    'Probability of death from non-communicable diseases (%)'
]
```

```
# Calculate the mean for each factor for males and females
means_by_sex = data[all_factors].groupby(data['Sex']).mean()
means_by_sex
```

```
[94]:      Controlled hypertension (%) \
Sex
Female      13.098798
Male        7.942564

      Hypertension among adults aged 30-79 years (%) \
Sex
Female      35.708919
Male        36.749932

      Raised Blood pressure (%)  Obesity prevalence among adults (%) \
Sex
Female      23.149276           16.003789
Male        26.330148           9.941222

      Overweight prevalence among adults(%) \
Sex
Female      40.535438
Male        36.178345

      Estimate of current cigarette smoking prevalence (%) \
Sex
Female      5.685149
Male        28.220607

      Alcohol Consumption  Receiving hypertension treatment (%) \
Sex
Female      2.041847           37.238591
Male        8.142684           23.704063

      Underweight prevalence among adults (%) \
Sex
Female      7.421808
Male        7.438059

      Probability of death from non-communicable diseases (%)
Sex
Female      18.413921
Male        25.438566
```

```
[120]: # Bar chart using means for males and females for each of the factors
# Grouping factors together to create separate plots.
hypertension_factors = [
```

```

    'Controlled hypertension (%)',
    'Hypertension among adults aged 30-79 years (%)',
    'Raised Blood pressure (%)',
    'Receiving hypertension treatment (%)']

weight_factors = [
    'Obesity prevalence among adults (%)',
    'Overweight prevalence among adults (%)',
    'Underweight prevalence among adults (%)']

lifestyle_factors = [
    'Estimate of current cigarette smoking prevalence (%)',
    'Alcohol Consumption', ]

death_factor = ['Probability of death from non-communicable diseases (%)']

# figure and axes
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# Bar width adjustment
bar_width = 0.35

# Function to plot bar charts for each category
def plot_bar_chart(factors, ax, title, means_by_sex):
    x_positions = np.arange(len(factors))

    ax.bar(x_positions - bar_width / 2, means_by_sex.loc['Male', factors],
           width=bar_width, color='darkorange', label='Male')

    ax.bar(x_positions + bar_width / 2, means_by_sex.loc['Female', factors],
           width=bar_width, color='grey', label='Female')

    ax.set_title(title)
    ax.set_ylabel('Mean Percentage')
    ax.set_xticks(x_positions)
    ax.set_xticklabels(factors, rotation=45, ha='right', fontsize=10)
    ax.legend()

# Plot Hypertension-related factors
plot_bar_chart(hypertension_factors, axes[0, 0], 'Hypertension-related
Factors', means_by_sex)

# Plot Weight-related factors
plot_bar_chart(weight_factors, axes[0, 1], 'Weight-related Factors',
means_by_sex)

# Plot Lifestyle-related factors

```

```

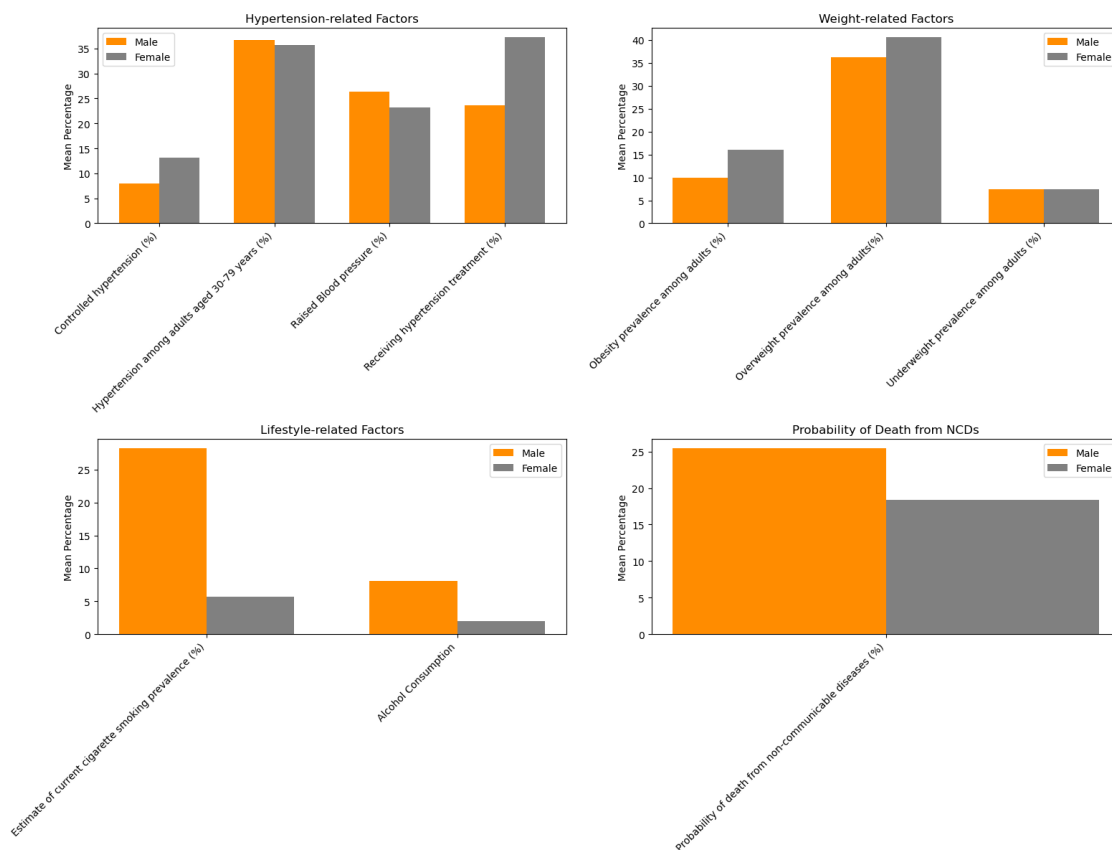
plot_bar_chart(lifestyle_factors, axes[1, 0], 'Lifestyle-related Factors',
               ↪means_by_sex)

# Plot Probability of Death
plot_bar_chart(death_factor, axes[1, 1], 'Probability of Death from NCDs',
               ↪means_by_sex)

# Adjust layout for better spacing
plt.tight_layout()

# Show the plot
plt.show()

```



Visual Analysis for bar charts

1.) Hypertension-related factors: The mean percentage of Hypertension and raised blood pressure is higher in males and yet the mean percentage receiving treatment for hypertension is lower in males than it is for females. Males also have lower controlled Hypertension on average than females which could potentially be a result of men receiving less treatment for hypertension. #####

2.) Weight-related factors: Obesity and overweight prevalence for females on average is

higher than for males. The overweight group has the highest prevalence out of the three weight groups shown here for both sexes. Underweight prevalence has the lowest prevalence for both sexes (7.5%). ##### 3.) Lifestyle-related factors: The mean estimate of current cigarette smoking prevalence is considerably higher in males than in females with 5% on average for females and 27.5% on average for males. Alcohol consumption is also higher in males with 2.5% on average for females and 7.5% on average for males. ##### 4.) Probability of death from Non-communicable diseases: Males have a higher probability of death from non-communicable diseases on average than females with a mean of 25% probability in males compared to 17.5% in females.

1.3 Statistical Analysis 1

1.3.1 How do lifestyle factors correlate with disease prevalence for males and for females?

Lifestyle factors: - cigarette smoking - alcohol consumption ##### Disease Prevalence: - hypertension-related factors - weight-related factors - mortality from non-communicable diseases

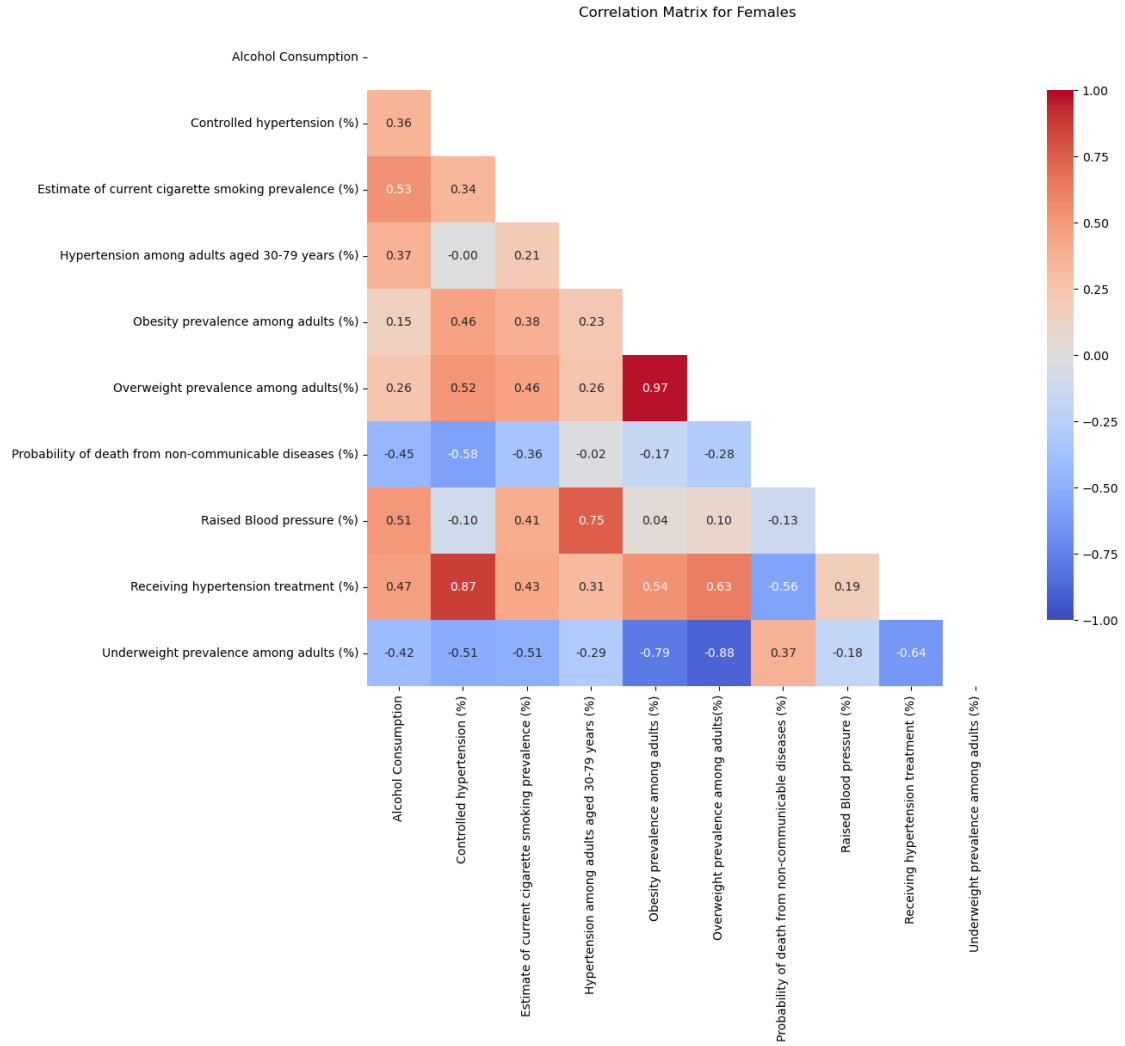
```
[123]: # Correlation matrices for males and females

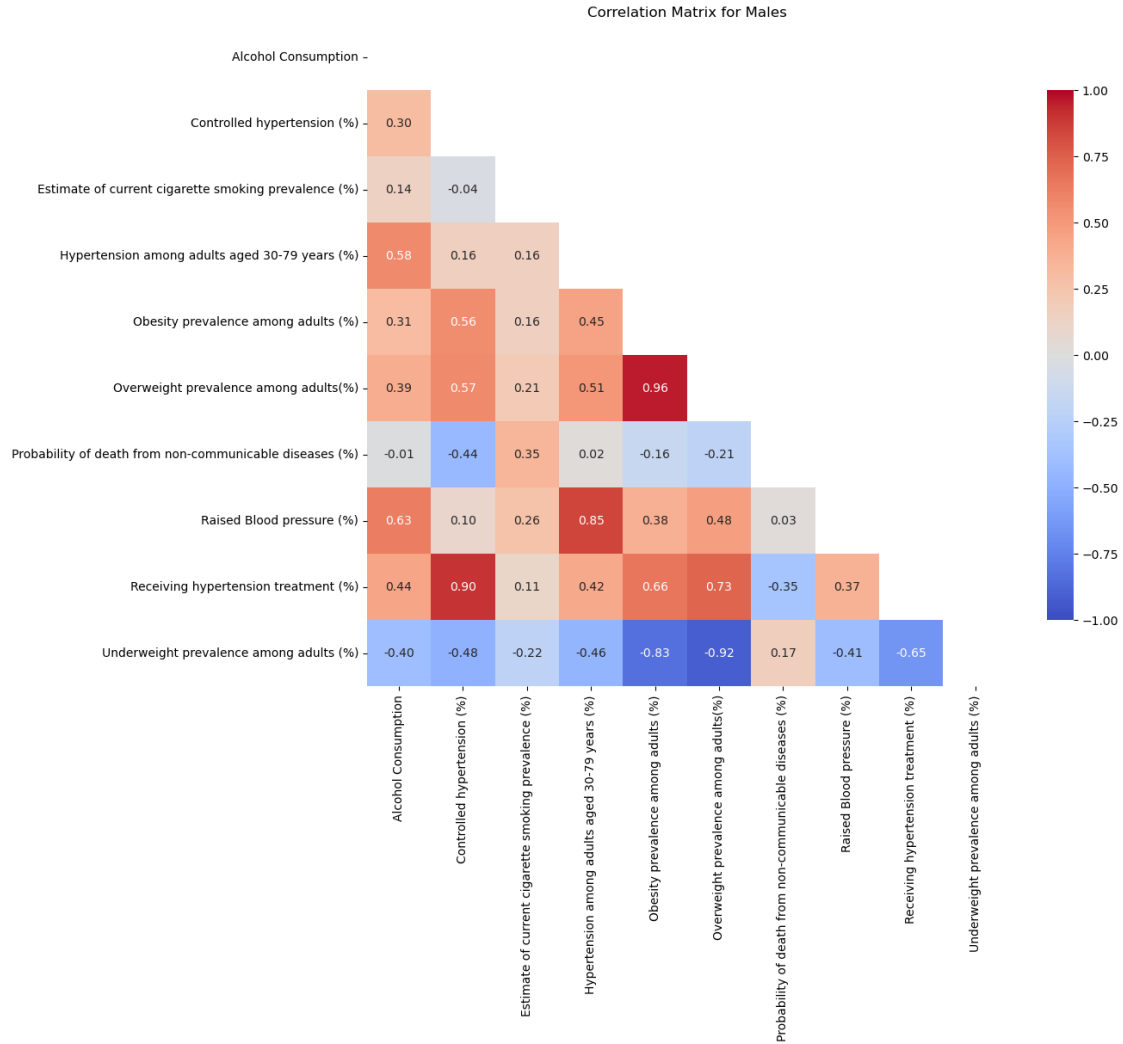
# Create correlation matrices
male_corr = male_numeric.corr()
female_corr = female_numeric.corr()

# For hiding repeats (top half of correlation matrix)
male_mask = np.triu(np.ones_like(male_corr, dtype=bool))
female_mask = np.triu(np.ones_like(female_corr, dtype=bool))

# Heatmap- females
plt.figure(figsize=(12, 10))
sns.heatmap(female_corr, annot=True, cmap='coolwarm', mask=female_mask,
            fmt=".2f", cbar_kws={"shrink": .8}, vmin=-1, vmax=1)
plt.title("Correlation Matrix for Females")
plt.show()

# Heatmap- male
plt.figure(figsize=(12, 10))
sns.heatmap(male_corr, annot=True, cmap='coolwarm', mask=male_mask,
            fmt=".2f", cbar_kws={"shrink": .8}, vmin=-1, vmax=1)
plt.title("Correlation Matrix for Males")
plt.show()
```



1.3.2 Correlation Analysis

Correlation matrix- females

1) For Alcohol consumption and disease prevalence:

- Alcohol consumption had a moderate positive correlation with raised blood pressure (0.51) and receiving hypertension treatment (0.47), slightly lower positive correlation between hypertension prevalence (0.37) and controlled hypertension (0.36).
- Overweight was weakly positively correlated with alcohol consumption (0.26) and obesity had an even lower correlation (0.15)
- Probability of death from NCBs was negatively correlated with alcohol consumption (-0.45). Underweight prevalence also had a negative correlation (-0.42).

2) For Smoking and disease prevalence:

- Smoking had moderate correlation with overweight prevalence (0.46), receiving hypertension treatment (0.43), raised blood pressure (0.41) and slightly lower positive correlation between smoking and obesity (0.38), controlled hypertension (0.34).
- Weak positive correlation between smoking and hypertension prevalence (0.21).
- Moderate negative correlation between underweight prevalence (-0.51) and smoking. And also between smoking and mortality.

Correlation matrix- males

1) For Alcohol consumption and disease prevalence:

- Moderate High positive correlation between Alcohol consumption and raised blood pressure (0.63)
- Alcohol consumption had a Moderate positive correlation with Hypertension prevalence (0.58), hypertension treatment (0.44), overweight prevalence (0.39), obesity prevalence and controlled hypertension (0.30).
- Underweight prevalence and Alcohol had a moderate negative correlation (-0.40).
- No correlation between Alcohol consumption and smoking (-0.01)

2) For Smoking and disease prevalence:

- Moderate to low correlation between smoking and mortality (0.35)
- Smoking prevalence had a weak positive correlation with raised blood pressure (0.26) and with overweight prevalence (0.21).
- little to no correlation with obesity prevalence (0.16), Hypertension prevalence (0.16) and Hypertension Treatment (0.11)

Key insights from both matrices

- The correlation analysis for females and males give very different results for most factors.
- Alcohol consumption and Hypertension factors positively correlated for both males and females.
- Smoking had a weak positive correlation with mortality for males and no correlation for females.

1.4 Statistical Analysis 2

Analysing the relationship between health behaviours and treatments and mortality risk from NCDs

- Our mortality risk data (probability of dying from non-communicable disease) is a percentage, a numerical continuous variable. We can therefore try to fit a linear regression model to our data with mortality risk as our outcome/dependent variable and the health factors as predictors.
- For our health behaviours we have some that are closely associated. For example overweight and obese, raised blood pressure and hypertension, and controlled hypertension and receiving

hypertension treatment. This is likely because these metrics measure very similar things so it would be pointless to include all of them in our model.

- To decide which ones to keep, we could use the correlation matrices to look at which of the factors has a stronger correlation with the probability of dying from NCDs.
- The ones with the highest correlation with mortality risk for both males and females are:
 - Raised Blood Pressure prevalence
 - Controlled Hypertension prevalence
 - Overweight prevalence

[]:

```
[126]: # importing the modules required for linear regression
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
```

```
[134]: # listing all predictor variables (too long to put directly in the model)
Predictors = [
    'Controlled hypertension (%)',
    'Overweight prevalence among adults(%)',
    'Estimate of current cigarette smoking prevalence (%)',
    'Alcohol Consumption',
    'Raised Blood pressure (%)',
    'Sex',
    'Underweight prevalence among adults (%)'
]
# dealing with the characters in the predictor names and 'Sex' which contains
↳ strings
formula = f"Q('{dependent_variable}') ~ " + " + ".join([f"Q('{col}')" for col
↳ in Predictors])

# Define the dependent variable (outcome)
dependent_variable = 'Probability of death from non-communicable diseases (%)'

# Linear regression model to relate mortality to health factors
m0 = sm.OLS.from_formula(formula, data=data)
r0 = m0.fit()
r0.summary()
```

[134]:

Dep. Variable:	Q('Probability of death from non-communicable diseases (%)')	R-squared:
Model:	OLS	Adj. R-squared:
Method:	Least Squares	F-statistic:
Date:	Wed, 08 Jan 2025	Prob (F-statistic):
Time:	22:25:26	Log-Likelihood:
No. Observations:	110310	AIC:
Df Residuals:	110302	BIC:
Df Model:	7	
Covariance Type:	nonrobust	

	coef	std err	t	P> t	[0
Intercept	21.2656	0.147	144.761	0.000	20
Q('Sex')[T.Male]	3.1438	0.102	30.784	0.000	2
Q('Controlled hypertension (%)')	-0.3555	0.002	-150.047	0.000	-0
Q('Overweight prevalence among adults(%))')	0.0546	0.002	26.528	0.000	0
Q('Estimate of current cigarette smoking prevalence (%)')	0.1170	0.004	31.260	0.000	0
Q('Alcohol Consumption')	-0.0198	0.009	-2.172	0.030	-0
Q('Raised Blood pressure (%)')	-0.0728	0.003	-24.758	0.000	-0
Q('Underweight prevalence among adults (%)')	0.0876	0.006	15.174	0.000	0
Omnibus:	39039.930	Durbin-Watson:	0.077		
Prob(Omnibus):	0.000	Jarque-Bera (JB):	176923.611		
Skew:	1.686	Prob(JB):	0.00		
Kurtosis:	8.207	Cond. No.	458.		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Interpreting the coefficients from the linear regression model All coefficients in this model are statistically significant at a 5% threshold.

Sex

- Males have a higher risk of mortality than females
- Males would have 3.1438% higher mortality risk % compared to the female.

Controlled Hypertension-

- for every 1% increase in controlled hypertension, there is a 0.3555% decrease in mortality risk %.

Raised Blood Pressure prevelance

- For every 1% increase in blood pressure, there is a 0.0728% decrease in mortality risk %.

Overweight prevalence among adults

- For every 1% increase in overweight prevalence, there is a 0.0546% increase in mortality risk %.

Estimate of current cigarette smoking prevalence

- For every 1% increase in overweight prevalence, there is a 0.0117% increase in mortality risk %.

Alcohol Consumption

- For every 1 litre increase in Alcohol consumption, there is a 0.0198% decrease in mortality risk %.

Underweight prevalence among adults

- For every 1% increase in underweight prevalence, there is a 0.0876% increase in mortality risk %.

Some Overall findings

- Out of all the health metrics here, controlling hypertension decreases the risk of mortality the most so controlling hypertension (through treatment, diet, exercise) may reduce mortality risks.
- Interestingly, according to our model, alcohol consumption also decreases mortality risk. The alcohol consumption category was one of the ones containing lots of missing values at the start which we imputed with the medians for the gender groups.
- Both underweight and overweight weight groups increased the mortality risk as did smoking.
- A strange finding from our model was that although controlled hypertension decreased hypertension the most, an increase in raised blood pressure also caused a slight decrease in mortality risk.

[]:

[]:

```
[139]: # Answering Some Questions using our model coefficients.
hypertension_coefficient = -0.3555
smoking_coefficient = 0.1170

# How would the probability of dying from an NCB change if the prevalence of
↳ Controlled Hypertension increased by 10%?
increase_hypertension_10 = hypertension_coefficient * 10
print(f'The mortality risk NCB if the prevalence of Controlled Hypertension
↳ increases by 10% would be {increase_hypertension_10:.2f} %')

# How would the probability of dying from an NCB change if the prevalence of
↳ Controlled Hypertension increased by 30%?
increase_hypertension_30 = hypertension_coefficient * 30
print(f'The mortality risk NCB if the prevalence of Controlled Hypertension
↳ increases by 10% would be {increase_hypertension_30:.2f} %')

# How would the probability of dying from an NCB change if the prevalence of
↳ Smoking decrease by 10%?
decrease_smoking_10 = smoking_coefficient * -10
print(f'The mortality risk NCB if the prevalence of Smoking decreases by 10%
↳ would be {decrease_smoking_10:.2f} %')
```

The mortality risk NCB if the prevalence of Controlled Hypertension increases by 10% would be -3.55 %

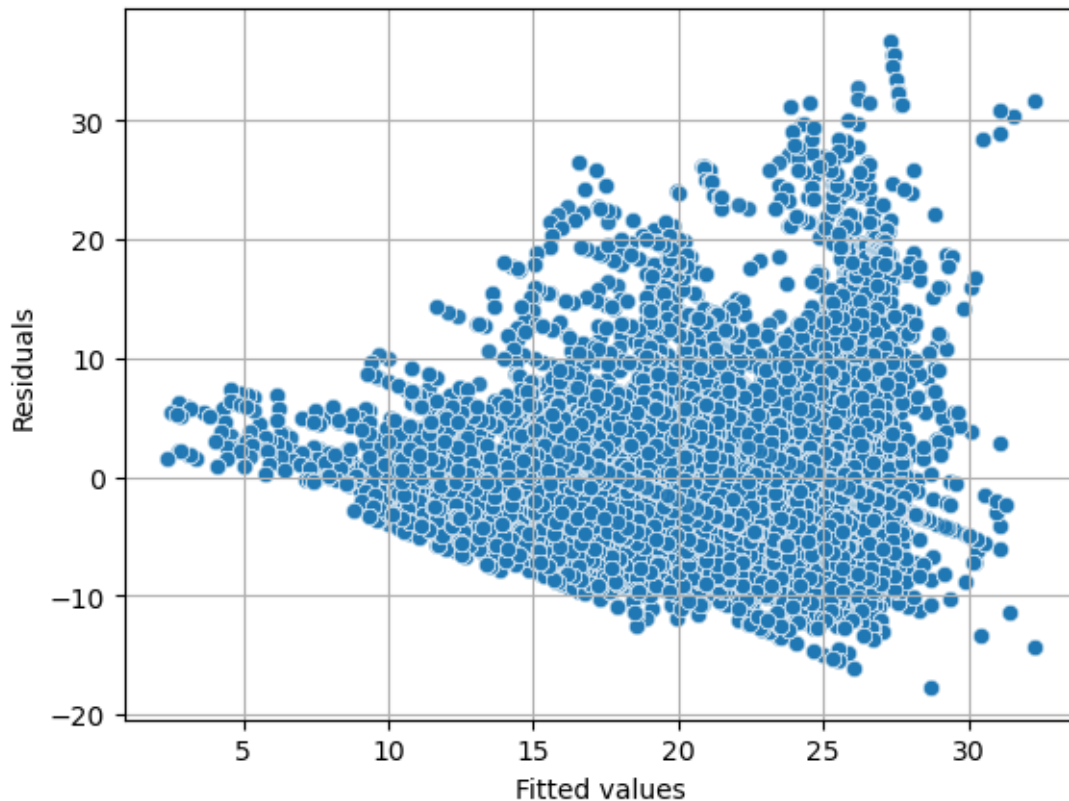
The mortality risk NCB if the prevalence of Controlled Hypertension increases by

10% would be -10.66 %

The mortality risk NCB if the prevalence of Smoking decreases by 10% would be -1.17 %

Checking the model fit is appropriate

```
[132]: # Checking assumption of linearity
pp = sns.scatterplot(x=r0.fittedvalues, y=r0.resid,alpha=0.5)
pp.grid(True)
pp.set_xlabel("Fitted values")
pp.set_ylabel("Residuals");
```



Analysis of residual plot The residuals are not scattered randomly. They appear to be following a funnel shape- deviations from the mean are higher at the right. This suggest non-linear data. The model for linear regression does not fit the data well since the assumption of linearity has been violated. This means that the estimates from our model are likely not accurate.

1.4.1 Conclusion- summary of findings and limitations of analysis

Findings

In our analysis we focused on using the global health data to help us look at overall gender disparities in health metrics. Looking at the bar charts for the hypertension-related factors we saw that males had an overall higher mean blood pressure than females but they also had lower controlled hypertension and did not receive treatment as much as females. Women also tended to be overweight or obese more than men. For lifestyle factors, males smoked and consumed alcohol more than females.

In the correlation matrix we saw that the Hypertension factors and Alcohol factors were positively correlated for both males and females. Both these factors were higher on average for males as we saw in the bar charts. This may perhaps be linked to the higher mortality for males. This could be studied further. Smoking had a very weak correlation with mortality for men and no correlation at all which is surprising considering smoking is often linked to non-communicable diseases, particularly for lungs. Smoking was however an estimate so it would be difficult to accurately collect data for this. Smoking was also a column that we used median imputation for cleaning the data for so this could have biased our results and affected our analysis as well.

Limitations

1) Handling of missing data

We had columns with large amounts of missing data and although we got rid of the ones with over 89.5% of the data missing when cleaning, there were some columns with lots of missing data that we could not remove so we did median imputation to deal with the missing values. One of these columns was the probability of dying from non-communicable diseases. This was essential for both statistics tasks 1 and 2 but it could have made our analysis biased as most of the values were just the median (by sex) of the rest of the column. This may have had a particularly negative impact on our linear regression model estimates.

2) The original data itself

For factors such as smoking and alcohol consumption, it may not be possible to get accurate data since people may not be honest and may not have even disclosed this (which may explain the missing data). Also, for global data like this, each country may have been using different methods to collect their data which may also mean our data may not be an accurate representation of these factors.

3) Regression model fit

Another issue was the linear regression itself. Looking at the residual plot analysis we can see that the assumption of linearity has been violated so the regression model is not a good fit for our data. This further questions the reliability of the estimates for the coefficients obtained from our analysis.

We therefore unfortunately cannot conclude anything with confidence from this analysis.

General Disclosure of AI: Chatgpt was used to aid with identifying errors in the code so that they could be fixed. In particular it was used to help generate the code for the boxplots and identify errors in the code whilst I was creating the bar charts.